

Ticket #004 - Threat Hunt Report

Executive Summary

A proactive threat hunt was conducted against the Ubuntu Linux lab system after unusual activity was reported without a confirmed indicator of compromise.

The hunt focused on the hypothesis that unauthorized persistence might be occurring through scheduled tasks such as cron jobs or systemd services.

The investigation identified two recently created cron jobs. One performed routine system-health monitoring. The second, a configuration backup job, initially appeared more suspicious because it executed every five minutes as root and stored archives containing system and SSH configuration files under /var/tmp.

Further investigation showed that both tasks were created by the analyst account through sudo during an interactive SSH session originating from 192.168.56.1. Historical login records showed repeated prior access by the analyst account from the same source address.

A hunt pivot identified an additional root cron task running every minute. Inspection showed that this task was a defensive SSH brute-force detection script.

Systemd services and timers were also reviewed. No unexplained persistence mechanism was identified.

Final Disposition

Close - Benign

Confidence

Medium

The evidence is most consistent with legitimate administrative and lab activity. No evidence was identified demonstrating account compromise, unauthorized persistence, malware execution, or activity requiring formal incident response.

Hunt Hypothesis

If unauthorized persistence through scheduled tasks is occurring on the Linux system, I would expect to observe newly created or modified cron jobs or systemd services associated with unusual commands, users, files, or execution times because scheduled tasks are commonly used by both legitimate administrators and attackers to execute commands persistently.

Known Facts

At the beginning of the hunt:

- A security analyst had observed unusual activity.
- No high-confidence security alert had triggered.
- No confirmed security incident had been declared.
- No specific IOC was available.
- The observed behavior had not been classified as legitimate or malicious.
- A proactive, hypothesis-driven threat hunt was requested.
- The target system was an authorized Ubuntu Linux lab host.

Unknowns

At the beginning of the investigation, it was unknown:

- What specific behavior caused concern.
- Which account performed the activity.
- Whether the activity was authorized.
- Whether persistence was involved.
- Whether cron or systemd had been modified.
- Whether privilege escalation had occurred.
- Whether the activity originated locally or remotely.
- Whether unusual scripts or processes were executing.
- Whether multiple systems were affected.

Assumptions

Initial assumptions included:

- Relevant logging remained available.
- Cron and systemd telemetry could be examined.
- Legitimate administrative activity was possible.
- Scheduled tasks were not automatically malicious.
- Recent configuration changes should leave observable evidence.
- The unusual activity might involve persistence, but alternative explanations remained possible.

These assumptions were reevaluated as evidence was collected.

Competing Explanations

1. Legitimate Administrative Activity

An administrator may have created scheduled tasks for monitoring, backups, maintenance, or automation.

2. Legitimate but Poorly Documented Configuration Change

A developer or administrator may have made an authorized change without formal documentation.

3. Unauthorized Persistence

An unauthorized actor may have created or modified a scheduled task to maintain continued execution.

4. Application or Software Automation

Installed software or another system component may have automatically created scheduled execution.

At least one benign explanation was maintained throughout the investigation to reduce confirmation bias.

Hunt Plan

The hunt initially focused on the following areas:

- Cron configuration
- Systemd services
- Systemd timers
- Authentication telemetry
- Sudo and privilege activity
- File metadata
- Script contents
- Process and execution evidence

The investigation would expand only when evidence created a new question or justified examining an area outside the original path.

Environment and Data Sources

System

- Host: ubuntu-soc-lab
- Operating System: Ubuntu Linux
- Primary investigative account: analyst

Data Sources

Cron Configuration

Examined because cron provides legitimate scheduled execution but can also provide persistence.

Systemd Services and Timers

Examined because systemd may provide persistent or scheduled execution.

Authentication Logs

Used to determine the origin of interactive sessions and identify unusual access.

Sudo Telemetry

Used to determine which account performed privileged changes.

File Metadata

Used to correlate modification times, ownership, permissions, and artifact creation.

Journal Logs

Used to confirm actual execution of scheduled commands.

Login History

Used to compare current access with previous analyst login behavior.

Investigation

Cron Enumeration

Cron configuration review identified two recently modified entries:

dev-health-check

- Executes every 15 minutes.
- Runs as root.
- Executes `/usr/local/bin/dev-health-check.sh`.
- Collects hostname, uptime, and filesystem usage.
- Writes to `/var/log/dev-health-check.log`.

This task had a clear administrative monitoring purpose.

config-backup

- Executes every five minutes.
- Runs as root.
- Executes `/usr/local/bin/config-backup.sh`.
- Archives `/etc/hosts` and `/etc/ssh/sshd_config`.
- Stores timestamped archives under `/var/tmp`.

This task was considered more suspicious because of its frequency, elevated privileges, and temporary-directory destination.

Artifact Investigation

The config-backup script created files such as:

`/var/tmp/dev-config-20260823-175433.tar.gz`

Artifacts were:

- Owned by root.
- Permission mode 600.
- Approximately 1.9 KB.

- Stored under /var/tmp.

Multiple archives confirmed that the scheduled task had executed repeatedly.

Journal telemetry independently confirmed cron execution at approximately:

- 18:00:01 UTC
- 18:05:01 UTC

This demonstrated actual scheduled execution rather than merely the presence of a cron configuration.

Authentication and Privilege Investigation

Sudo telemetry showed that the analyst account:

- Created /usr/local/bin/dev-health-check.sh.
- Made the script executable.
- Created /etc/cron.d/dev-health-check.
- Created /usr/local/bin/config-backup.sh.
- Made the script executable.
- Manually executed config-backup.sh.
- Created /etc/cron.d/config-backup.
- Set permissions on both cron configuration files.

One sudo authentication failure occurred at 17:50:04 UTC.

This was followed approximately five seconds later by successful sudo activity from the same account and session.

The isolated failure did not materially support an unauthorized-access hypothesis.

Session Origin Investigation

The active analyst pts/0 session originated from:

192.168.56.1

SSH telemetry showed successful authentication for analyst from that address at:

17:35:59 UTC

Historical login records showed repeated analyst sessions from the same address on numerous prior dates.

Interpretation

The session origin was consistent with historical access patterns.

This weakened the explanation that an unknown remote account established the recently identified scheduled tasks.

Historical consistency, however, was treated as supporting evidence rather than proof of formal authorization.

Hunt Pivot

What Was Discovered

While reviewing cron execution telemetry, an unexpected script was observed running every minute:

```
/usr/local/bin/detect-ssh-bruteforce.sh
```

This task had not appeared in the initial `/etc/cron.d` enumeration.

Why It Mattered

The task executed as root every minute and therefore represented an additional persistent scheduled execution mechanism.

Its existence required investigation rather than dismissal based on its filename.

New Question

Was the task another unauthorized persistence mechanism, or was it a legitimate defensive control?

Additional Evidence

The script:

- Examines recent SSH journal events.
- Searches for failed password attempts.
- Counts failures by source IP.
- Uses a threshold of five failures within five minutes.
- Generates an auth warning when the threshold is reached.

The file timestamp was July 20, 2026.

Review of the root crontab showed:

```
* * * * * /usr/local/bin/detect-ssh-bruteforce.sh
```

This explained the repeated execution observed in the journal.

Pivot Assessment

The pivot weakened the original hypothesis.

The task was consistent with defensive SSH brute-force monitoring rather than unauthorized persistence.

The pivot demonstrated that persistent execution must be evaluated in context rather than classified solely on frequency, privileges, or scheduling mechanism.

Systemd Review

Systemd timers reviewed included common Ubuntu maintenance functions such as:

- apt-daily
- apt-daily-upgrade
- logrotate
- fstrim
- sysstat
- dpkg-db-backup
- fwupd-refresh
- systemd-tmpfiles-clean

No suspicious newly created timer was identified.

One recently modified service was reviewed:

```
/etc/systemd/system/company-web.service
```

The service:

- Was created before the current hunt.
- Is described as an Internal Company Web Application.
- Runs as the analyst user.

- Executes Python application code from /home/analyst/internal-web-outage-lab/app.py.
- Was active and operating normally.
- Served the application locally on 127.0.0.1:5050.

The service appeared consistent with legitimate lab application hosting.

Evidence Classification

Facts

- The analyst account authenticated through SSH from 192.168.56.1.
- The analyst account used sudo to create both newly identified cron tasks.
- config-backup.sh archives /etc/hosts and /etc/ssh/sshd_config.
- The backup archives were written under /var/tmp.
- Cron executed config-backup.sh automatically.
- The root crontab runs detect-ssh-bruteforce.sh every minute.
- The SSH detection script checks failed authentication attempts.
- Historical analyst logins repeatedly originated from 192.168.56.1.
- No new suspicious systemd timer was identified.

Inferences

- The dev-health-check task is consistent with legitimate monitoring.
- The config-backup activity is most consistent with administrative or lab activity.
- The detect-ssh-bruteforce task is likely a defensive security control.
- The company-web service is consistent with a legitimate internal lab application.
- The investigated activity does not currently justify incident escalation.

Assumptions

- The analyst account activity was formally authorized.
- Historical source consistency indicates legitimate ownership of the session.
- The company-web application was formally approved.

These assumptions could not be independently confirmed through change-management records.

Findings

The investigation established that scheduled execution mechanisms were present, including newly created cron jobs.

One task initially displayed several characteristics that warranted investigation:

- Root execution
- Five-minute schedule
- SSH configuration collection
- Temporary-directory storage

However, additional context significantly changed the assessment.

The task was created by the analyst account during an interactive SSH session whose source matched historical account activity.

The investigation also identified an unexpected root cron task, but the task was found to perform defensive SSH monitoring.

Systemd review did not identify additional unexplained persistence.

No collected evidence demonstrated:

- Account compromise
- Unauthorized login
- Malware
- External command and control
- Data exfiltration
- Credential theft
- Hidden malicious systemd persistence
- Unauthorized user creation
- Security-control disabling

Disposition

Close - Benign

The evidence most strongly supports legitimate administrative and lab activity.

The investigation identified unusual behavior worth examining, but correlation across authentication, sudo, scheduled execution, file metadata, script contents, and historical login behavior provided a reasonable benign explanation.

Escalation is not recommended at this time.

Confidence Assessment

Confidence: Medium

Medium confidence was selected because multiple independent telemetry sources support the same conclusion.

Strongest Supporting Evidence

The strongest evidence was the correlation between:

- The analyst SSH session.
- The repeated historical use of 192.168.56.1.
- The sudo commands that created the scheduled tasks.
- The actual contents and behavior of the scripts.
- Cron execution telemetry.

Missing Evidence

The primary missing evidence is independent confirmation that the changes were formally approved.

No change ticket, administrator approval record, or centralized enterprise identity record was available.

Evidence That Could Increase Confidence

Confidence could increase to High if:

- A change-management record confirmed the task creation.
- An administrator independently confirmed authorization.
- Asset-management records confirmed 192.168.56.1 as the expected administrative source.
- Centralized identity telemetry confirmed the session.

Evidence That Could Change the Conclusion

The conclusion would require reevaluation if new evidence showed:

- Account compromise.
- An unexpected login source.

- Stolen credentials.
- Hidden persistence.
- External network communication associated with the task.
- Unauthorized access to the archived files.
- Malicious modification of scheduled scripts.

Limitations

The hunt was conducted in a controlled lab environment.

Limitations included:

- No centralized SIEM.
- No enterprise EDR telemetry.
- No formal change-management database.
- No identity-provider telemetry.
- No authoritative asset ownership records.
- Limited historical process telemetry.
- No enterprise-wide search across additional hosts.

These limitations prevented independent confirmation of formal authorization.

Recommendations

1. Maintain centralized logging for Linux authentication, sudo, cron, and systemd activity.
2. Implement file integrity monitoring for:
 - /etc/cron*
 - root user crontabs
 - /etc/systemd/system
 - /usr/local/bin
3. Establish approved baselines for scheduled tasks.
4. Alert on newly created root-level cron jobs and systemd services.
5. Correlate administrative changes with formal change-management records.
6. Monitor scripts executing from temporary directories.

7. Retain process execution telemetry to establish parent-child relationships.
8. Document defensive automation such as detect-ssh-bruteforce.sh so future analysts can quickly determine its purpose.
9. Periodically review root crontabs in addition to /etc/cron.d.
10. Avoid classifying persistence mechanisms as malicious without evaluating account, command, file, network, and historical context.

Stopping Criteria

The hunt stopped after:

- The original hypothesis was tested.
- Cron configuration was reviewed.
- Actual cron execution was confirmed.
- Referenced scripts were inspected.
- File artifacts were analyzed.
- Sudo activity was correlated.
- Authentication telemetry was reviewed.
- Session origin was established.
- Historical access behavior was compared.
- The required hunt pivot was completed.
- Root crontab activity was investigated.
- Systemd services and timers were reviewed.
- Competing explanations were evaluated.

At that point, no remaining evidence pointed toward another meaningful investigative lead.

Although additional logs or systems could always be examined, further investigation was unlikely to materially change the disposition.

Continuing to collect data without a specific unanswered question would not have been proportionate or hypothesis-driven.

Final Assessment

Disposition: Close - Benign

Confidence: Medium

The available evidence does not support escalation to formal incident response.

The hunt demonstrated that suspicious characteristics such as root execution, frequent scheduling, temporary storage, and persistence must be interpreted in context.

The strongest conclusion supported by the evidence is that the observed activity was consistent with legitimate administrative and lab behavior.